



Universidad Tecnológica de Durango

Tecnologías de la Información

Probabilidad y Estadística

Statistical Analysis

Statistical Analysis of Qualitative and Quantitative Variables

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Objective of the analysis

Survey students to correlate the amount of personal study time dedicated per week with the number of failed subjects, in order to analyze the impact of study hours on academic performance by degree program among UTD students.

Population:

- Finite (UTD students)

Qualitative Data:

- Degree Program
- Current Semester

Quantitative Data:

- Study Hours
- Failed Subjects

Sampling:

- Approximately 100 students

Qualitative Data:

- Current Degree Program
- Current Semester

Quantitative Questions:

- How many subjects have you failed during this semester?
- How many hours per week do you dedicate to studying those subjects?

Data analysis

Descriptive statistics

Frequency tables - qualitative

Frequency tables are used to check the stable statistics between data on qualitative data (in this case we will use the 4-month period on course and the career coursing)

Frecuencias de Carrera

Var1	Freq
Desarrollo de negocios	35
Diseño digital	13
Lengua Inglesa	4
Logística	19
Mantenimiento industrial	3
Mecatrónica	13
Tecnologías de la información	15

Frecuencias de Cuatrimestre

Var1	Freq
3	32
5	29
8	39
11	2

Frequency tables – quantitative

Frequency tables are used mainly to check the stable statistics between data on quantitative data (in this case we will use the time studying and the failed classes)

Tabla de frecuencia - Materias reprobadas

lim_inf	lim_sup	marca	frecuencia	frec_rel
0	1	0.5	40	0.3922
1	2	1.5	32	0.3137
2	3	2.5	16	0.1569
3	4	3.5	8	0.0784
4	5	4.5	2	0.0196
5	6	5.5	1	0.0098
6	7	6.5	1	0.0098
7	8	7.5	1	0.0098
8	9	8.5	0	0.0000
9	10	9.5	1	0.0098

Tabla de frecuencia - Materias reprobadas

lim_inf	lim_sup	marca	frecuencia	frec_rel
0	5	2.5	72	0.7059
5	10	7.5	22	0.2157
10	15	12.5	6	0.0588
15	20	17.5	2	0.0196

Statistical measures

This relies on the data with: mean, medium, mode, range, variance, standard deviation, quartile, decyl and percentile over quantitative data

```
## Failed classes
```

```
## $media
```

```
## [1] 1.215686
```

```
##
```

```
## $mediana
```

```
## [1] 1
```

```
##
```

```
## $moda
```

```
## [1] 0
```

```
##
```

```
## $rango
```

```
## [1] 10
```

```
##
```

```
## $varianza
```

```
## [1] 2.56688
```

```
##
```

```
## $desviación
```

```
## [1] 1.602149
```

```
##
```

```
## $cuartiles
```

```
## 25% 50% 100%
```

```
## 0 1 10
```

```
##
```

```
## $deciles
```

```
## 10% 20% 30% 40% 50% 60% 70% 80% 90% 100%
```

```
## 0 0 0 1 1 1 1 2 3 10
```

```
##
```

```
## $percentiles
```

```
## 1% 2% 3% 4% 5% 6% 7% 8% 9% 10% 11% 12
```

```
% 13%
```

```
## 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.0
```

```
0 0.00
```

```
## 14% 15% 16% 17% 18% 19% 20% 21% 22% 23% 24% 25
```

```
% 26%
```

```
## 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.0
```

```
0 0.00
```

```
## 27% 28% 29% 30% 31% 32% 33% 34% 35% 36% 37% 38
```

```
% 39%
```

```
## 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.0
```

```
0 0.39
```

```
## 40% 41% 42% 43% 44% 45% 46% 47% 48% 49% 50% 51
```

```
% 52%
```

```
## 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.0
```

```
0 1.00
```

```
## 53% 54% 55% 56% 57% 58% 59% 60% 61% 62% 63% 64
```

```
% 65%
```

```
## 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.0
0 1.00
## 66% 67% 68% 69% 70% 71% 72% 73% 74% 75% 76% 77
% 78%
## 1.00 1.00 1.00 1.00 1.00 1.71 2.00 2.00 2.00 2.00 2.00 2.0
0 2.00
## 79% 80% 81% 82% 83% 84% 85% 86% 87% 88% 89% 90
% 91%
## 2.00 2.00 2.00 2.00 2.00 2.00 2.00 2.00 2.87 3.00 3.00 3.0
0 3.00
## 92% 93% 94% 95% 96% 97% 98% 99% 100%
## 3.00 3.00 3.00 3.95 4.00 4.97 5.98 6.99 10.00

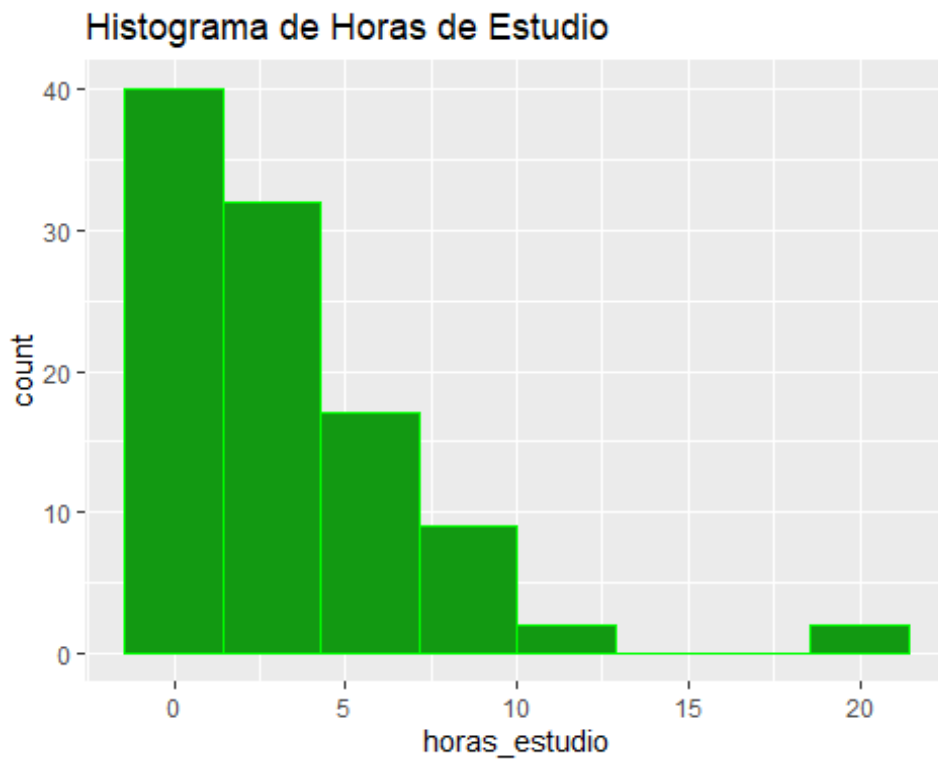
##
##
##
## Time studying

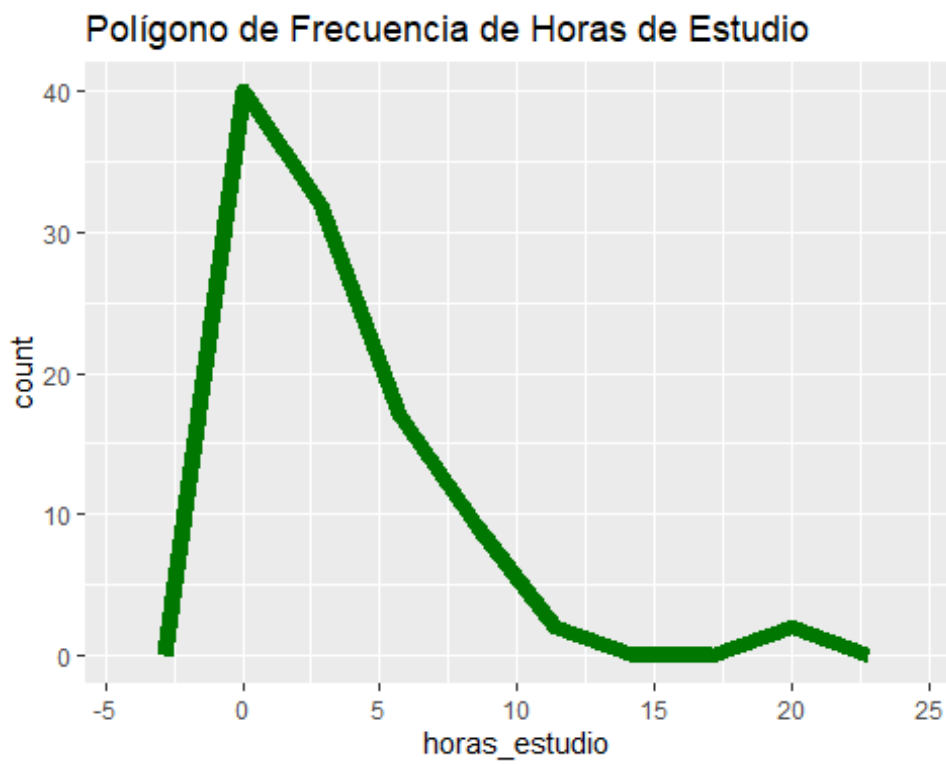
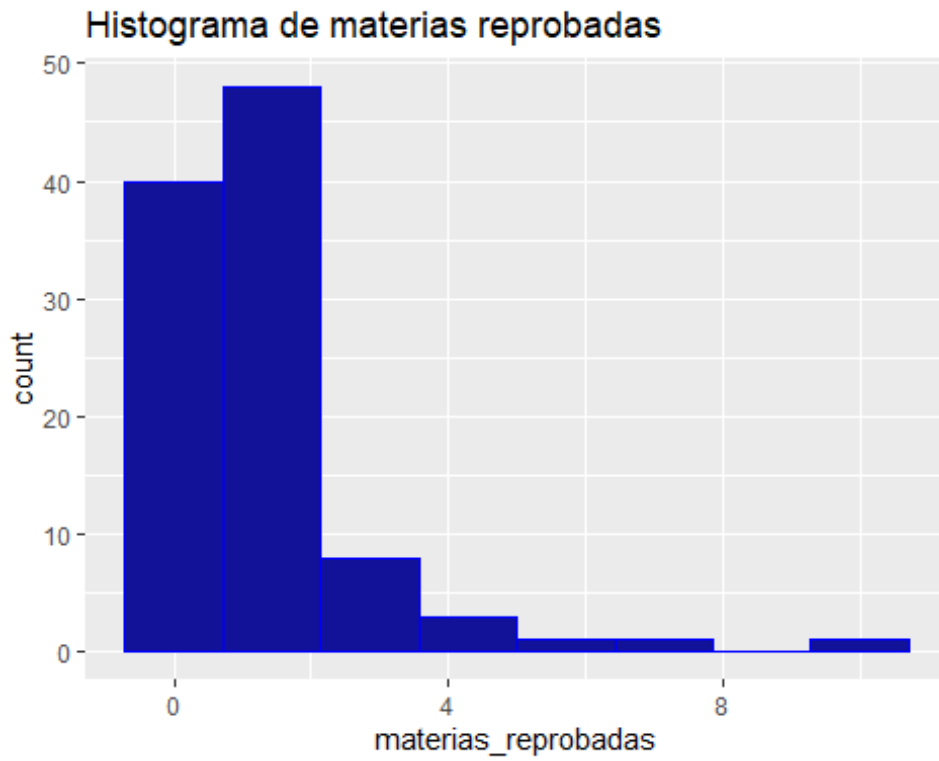
## $media
## [1] 3.254902
##
## $mediana
## [1] 2
##
## $moda
## [1] 0
##
## $rango
## [1] 20
##
## $varianza
## [1] 14.90468
##
## $desviación
## [1] 3.860658
##
## $cuartiles
## 25% 50% 100%
## 0 2 20
##
## $deciles
## 10% 20% 30% 40% 50% 60% 70% 80% 90% 100%
## 0 0 1 2 2 3 4 5 8 20
##
## $percentiles
## 1% 2% 3% 4% 5% 6% 7% 8% 9% 10% 11% 12
% 13%
```

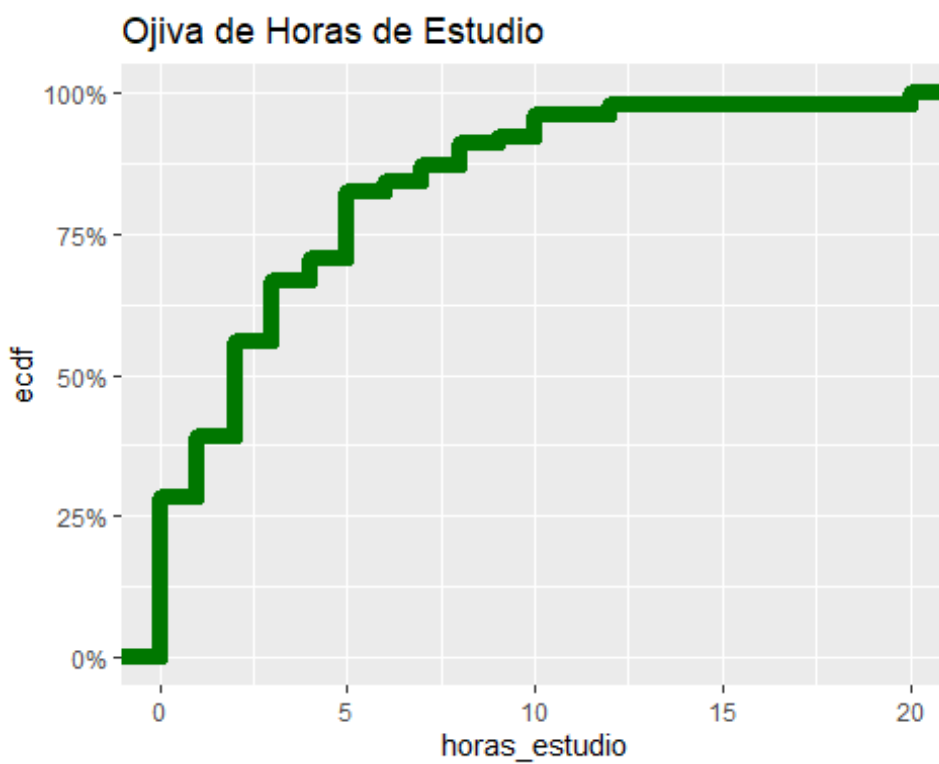
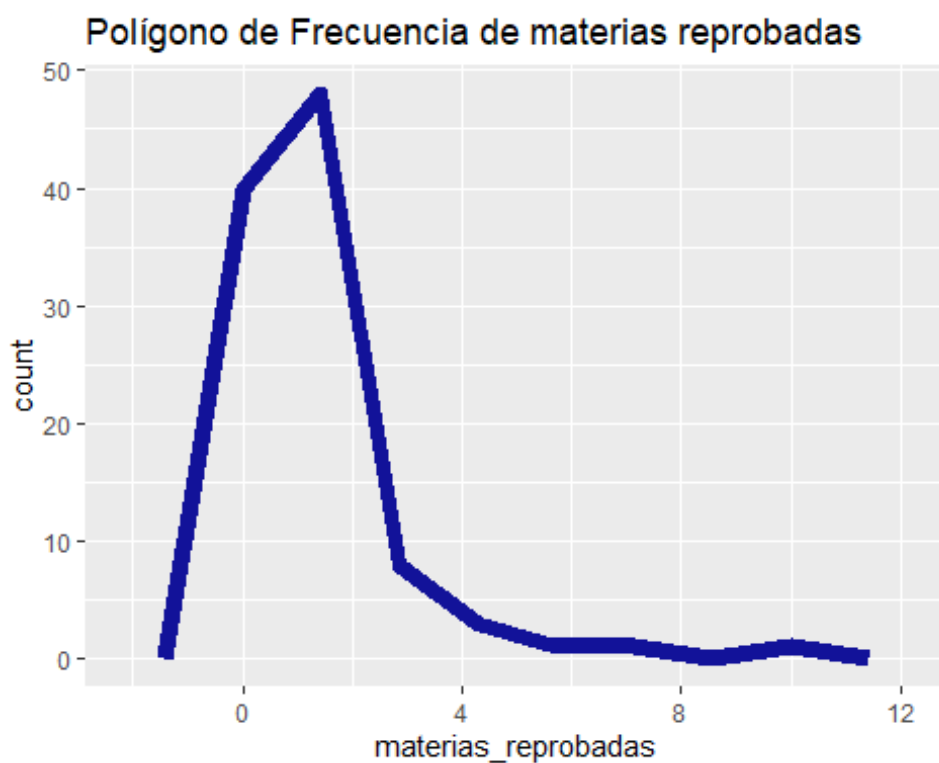
##	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.0
0	0.00											
##	14%	15%	16%	17%	18%	19%	20%	21%	22%	23%	24%	25
%	26%											
##	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.0
0	0.00											
##	27%	28%	29%	30%	31%	32%	33%	34%	35%	36%	37%	38
%	39%											
##	0.00	0.28	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.0
0	1.39											
##	40%	41%	42%	43%	44%	45%	46%	47%	48%	49%	50%	51
%	52%											
##	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.0
0	2.00											
##	53%	54%	55%	56%	57%	58%	59%	60%	61%	62%	63%	64
%	65%											
##	2.00	2.00	2.00	2.56	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.0
0	3.00											
##	66%	67%	68%	69%	70%	71%	72%	73%	74%	75%	76%	77
%	78%											
##	3.00	3.67	4.00	4.00	4.00	4.71	5.00	5.00	5.00	5.00	5.00	5.0
0	5.00											
##	79%	80%	81%	82%	83%	84%	85%	86%	87%	88%	89%	90
%	91%											
##	5.00	5.00	5.00	5.00	5.83	6.00	6.85	7.00	7.00	7.88	8.00	8.0
0	8.00											
##	92%	93%	94%	95%	96%	97%	98%	99%	100%			
##	8.92	9.93	10.00	10.00	10.00	11.94	12.00	19.92	20.00			

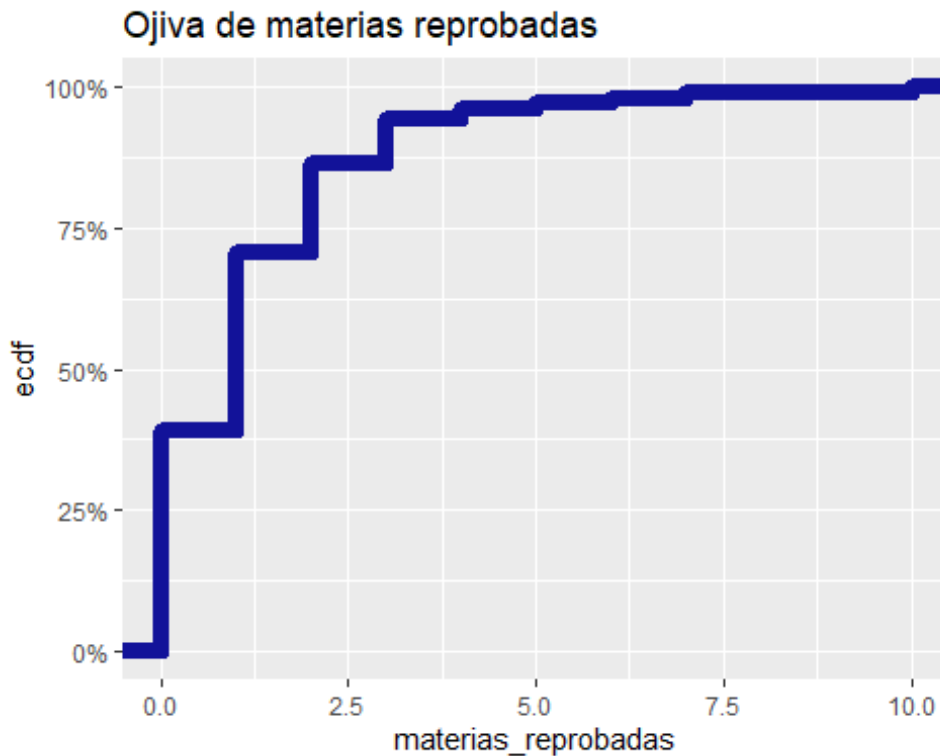
Graphs

In this scenario, we will use the quantitative data to Histogram, Frequency polygon and Ogives









With this graphs, we can conclude some information: - A lot of people struggle on at least 1 class over people who pass clean at the moment of the data recollection - Some people tries to study more, while almost half the people doesn't have any time to study - On both graphs, almost 75% of people are on the low spectrum between the time to study (little to no time to study) and the failed classes (little to no worries on classes)

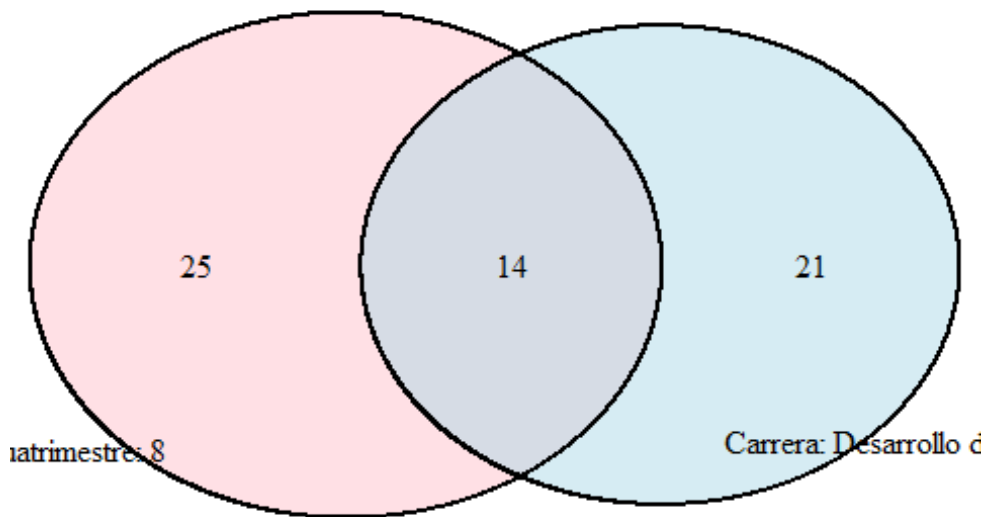
Statistical analysis

For these examples we will use 2 data

- 1.- People who are on DN
- 2.- People on the eighth 4-month period

Venn diagram

This will show us how much people are inside DN and are on their 8th 4-month period



```
## (polygon[GRID.polygon.228], polygon[GRID.polygon.229], polygon[GRID.polygon.230], polygon[GRID.polygon.231], text[GRID.text.232], text[GRID.text.233], text[GRID.text.234], text[GRID.text.235], text[GRID.text.236])
## [1] 0.3431373 0.3823529 0.1372549 0.3589744
```

Intersection, Union and Difference

With the Venn-Euler diagram, we can see that:

```
## The union between both states are: 60
## The intersection between both states is:
## 14
## The differences between both states are:
## 21 (a-b)
## and
## 25 (b-a)
```

Probability calculation

If we were to pick a random person inside the data...

```
## You will pick a DN student at 34.31373 %
```

```
## You will pick a eighth 4-month period student at a: 38.23529 %
```

Joint and conditional probability

```
## Probability that a person is both a DN student and a eighth 4-month pe  
riod student: 13.72549 %
```

```
## Total people who is at eighth given are at DN: 40 %
```

```
## Total people who is at DN given they are at eighth: 35.89744 %
```

Hypothesis test

Hypothesis

Null hypothesis (H0): There are no correlation between time studying and the failed classes
Alt hypothesis (H1): There are some correlations between the time studying and the failed classes.

```
##  
## Welch Two Sample t-test  
##  
## data: grupo0 and grupo1  
## t = 0.37504, df = 64.443, p-value = 0.7089  
## alternative hypothesis: true difference in means is not equal to 0  
## 95 percent confidence interval:  
## -1.276613 1.866809  
## sample estimates:  
## mean of x mean of y  
## 3.550000 3.254902
```

Based on the students' t-test we can say that data is far more random than expected and that time studying does not affect the failed classes (p value is over 0.05)

There is another test we can use.

```
##
## Wilcoxon rank sum test with continuity correction
##
## data: grupo0 and grupo1
## W = 1238, p-value = 0.9917
## alternative hypothesis: true location shift is not equal to 0
```

In this one, the Wilcoxon rank test shows that there is almost no correlation between data.

Scatter Plots and Correlation Analysis

Correlation value

```
##
## Pearson's product-moment correlation
##
## data: df_cor$materias_reprobadas and df_cor$horas_estudio
## t = 0.32659, df = 100, p-value = 0.7447
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## -0.1628667 0.2256843
## sample estimates:
## cor
## 0.03264207

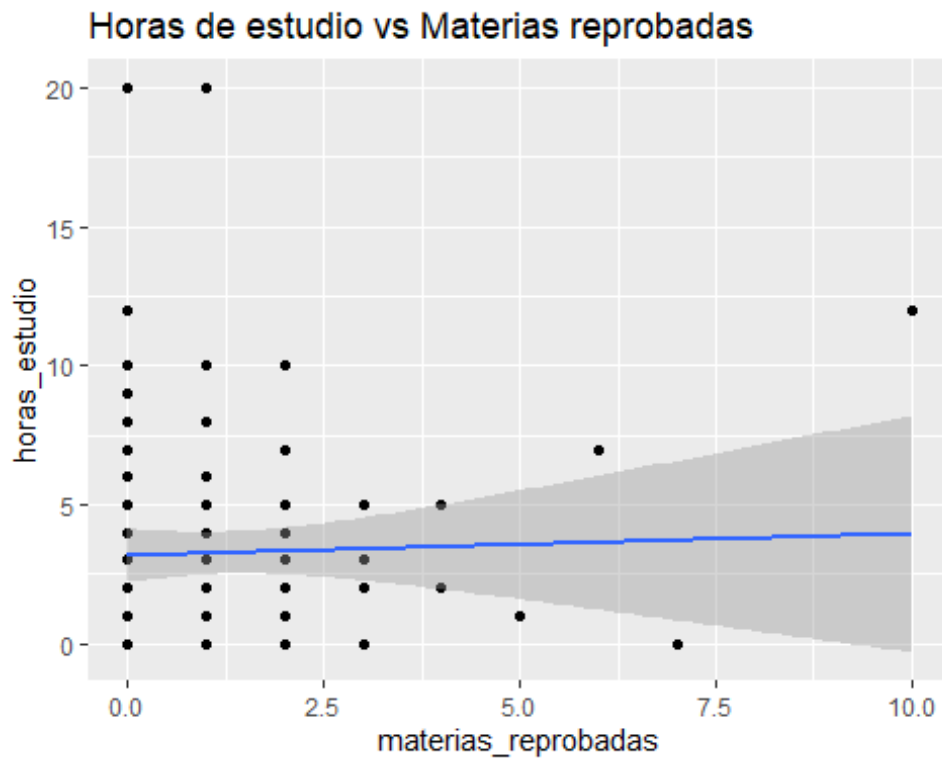
##
## Call:
## lm(formula = horas_estudio ~ materias_reprobadas, data = df_cor)
##
## Residuals:
## Min 1Q Median 3Q Max
## -3.710 -3.159 -1.238 1.762 16.841
##
## Coefficients:
## Estimate Std. Error t value Pr(>|t|)
## (Intercept) 3.15928 0.48286 6.543 2.6e-09 ***
## materias_reprobadas 0.07866 0.24084 0.327 0.745
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.878 on 100 degrees of freedom
```

```
## Multiple R-squared:  0.001066,   Adjusted R-squared:  -0.008924
## F-statistic: 0.1067 on 1 and 100 DF,  p-value: 0.7447

## The correlation between the studying time and the failed assignments
is for: 0.03264207
```

Scatter plot visualization.

```
## `geom_smooth()` using formula = 'y ~ x'
```



Conclusions

With this analysis we could see that, based on the population of 103 people inside the university:

- There are a lot of people studying DN.
- There are equal amounts of people studying on third, fifth and eighth 4-month periods inside the university.
- A lot of people either don't study at home or have little to no problem inside their classes.
- There are no little to no correlations between the time studying outside school and the grades shown on picture.

At least with the population we managed to interview, we could not determine if there existed any correlation between the time studying and the failed classes. Even if the tests say that there is no correlation, it is important to try this test on a much bigger population.

Statistics makes us learn a lot about what we can count inside something we consider important with way more precision and in way less time.

Taking into consideration what the numbers say might be something you want to get, this makes us get into smart decisions about what we can do to satisfy someone/get better at counting/identifying people/etc.

Using statistics day by day can improve your productivity, save time and making sure you are making a good decision.